Three Exact Results Arising from Collatz-Type Dynamics: Valuation Equidistribution, Fermat Bases, and a Transcendental Density Constant

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Abstract

The study of the accelerated Collatz map $\mathrm{step}(x) = (3x+1)/2^{\mathrm{v}_2(3x+1)}$ throws off questions that are not about the Collatz conjecture at all. We isolate three of them and settle them completely. *(i)* The valuations produced by the accelerated step obey an exact 2-adic law: for every odd base a, $\mathrm{v}_2(an+1) \geq k \iff n \equiv -a^{-1} \pmod{2^k}$, so $\mathrm{v}_2(an+1)$ is Geom(1/2) with mean 2 and variance 2, independently of a; and the two valuations $\mathrm{v}_2(3n\pm1)$ of the two-sided step are anti-correlated with covariance exactly -1. *(ii)* The map $x \mapsto ax+1$ (a odd) admits closed-form polynomial orbits if and only if $a-1$ is a power of 2, i.e. $a = 2^e+1$ is a Fermat-type base. *(iii)* The density constant $C_{\mathrm{Collatz}} = \frac{1}{2} - \sum_{n\geq 2} 2^{-\lfloor n \log_2 3 \rfloor} = 0.2862745023241087\ldots$ has a Sturmian binary expansion and is therefore transcendental; so is $C_a = 1 - \sum_{n\geq 1} 2^{-\lfloor n \log_2 a \rfloor}$ whenever $\log_2 a$ is irrational. Result (i), and the orbit formula of (ii), are formalised in Lean 4 and machine-checked with kernel-only axioms; (iii) is verified by exact computation and follows from known transcendence criteria. **None of the three depends on the Collatz conjecture, and none is claimed to advance it.** We also record, with explicit witnesses, a limitative fact that is easy to get wrong experimentally: for no modulus M is $\mathrm{v}_2(3x+1)$ a function of x mod M, so no finite automaton on residues can model the dynamics.

Keywords: Collatz map; 2-adic valuation; equidistribution; Sturmian words; transcendence; Beatty sequences; formalised mathematics; Lean 4.
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1 Introduction

For odd x the *accelerated* (Syracuse) Collatz step is

$$\mathrm{step}(x) = \frac{3x+1}{2^{\mathrm{v}_2(3x+1)}}, \tag{1}$$

where v_2 denotes the 2-adic valuation. Whether every orbit reaches 1 is the Collatz conjecture, open since Collatz's 1937 formulation; see Lagarias [4, 5] for the standard survey, Terras [11] for the classical density result, and Tao [10] for the strongest current "almost all" theorem. **This paper does not address the conjecture.**

What it does address is a cluster of questions that arise while studying (1) but whose statements never mention it. Each is settled here, and each stands whether or not the conjecture is true:

(i) an exact equidistribution law for the valuations $\mathrm{v}_2(3^r m - 1)$ (Section 2);
(ii) a characterisation of the bases a for which $ax+1$ has closed-form orbits (Section 3);
(iii) the transcendence of a density constant attached to the map (Section 4).

Method, and why this is experimental mathematics. All three began as computational observations and were then either proved outright or reduced to a known criterion. Two of the three are additionally *formalised in Lean 4* and checked by its kernel with no `sorry` and no custom axioms, which we regard as the appropriate standard of evidence when the source of a conjecture is a machine. Section 6 documents the verification, and Section 5 records — with explicit counterexamples — a false reduction that is easy to reach experimentally and that we believe deserves to be stated plainly in the literature.

2 An exact equidistribution law

Throughout this section m ranges over the odd residues modulo 2^k, of which there are 2^{k-1}.

Lemma 2.1 (Invertibility)**.** *3 is a unit in $\mathbb{Z}/2^k\mathbb{Z}$ for every $k \geq 1$; hence 3^r is a unit and $m \mapsto 3^r m$ permutes $(\mathbb{Z}/2^k\mathbb{Z})^\times$.*

Proof. $\gcd(3, 2) = 1$, so 3 is invertible modulo every power of 2. Units form a group, so 3^r is a unit and multiplication by it is a bijection of the unit group. □

Theorem 2.2 (Exact equidistribution)**.** *Let $r, k \geq 1$. There is* exactly one *odd $m \in [0, 2^k)$ with $\mathrm{v}_2(3^r m - 1) \geq k$.*

Proof. $\mathrm{v}_2(3^r m - 1) \geq k$ means $3^r m \equiv 1 \pmod{2^k}$, i.e. $m \equiv (3^r)^{-1} \pmod{2^k}$. By Lemma 2.1 this inverse exists, is unique modulo 2^k, and is odd (being a unit). Hence exactly one odd residue satisfies the condition. □

Corollary 2.3 (Geometric law)**.** *Let m be uniform on the odd residues modulo 2^k. For $1 \leq j < k$,*

$$\Pr\left[\mathrm{v}_2(3^r m - 1) = j\right] = 2^{-j},$$

independently of r, and $\Pr[\mathrm{v}_2 \geq k] = 2^{-(k-1)}$. Letting $k \to \infty$, the valuation is Geom(1/2) *on $\{1, 2, \dots\}$ with $\mathbb{E}[\mathrm{v}_2] = \sum_{j \geq 1} j 2^{-j} = 2$.*

Proof. By Theorem 2.2 the odd residues with $\mathrm{v}_2 \geq j$ form a single class modulo 2^j, hence a fraction $2^{-(j-1)}$ of the odd residues; likewise $\Pr[\mathrm{v}_2 \geq j + 1] = 2^{-j}$. Subtract. Independence of r is exactly Lemma 2.1. □

Remark 2.4 (Scope of the base)**.** The proof uses only that the base is odd (hence a 2-adic unit) and that units form a group. So for every odd a: for all $r, k \geq 1$ exactly one odd $m \in [0, 2^k)$ has $\mathrm{v}_2(a^r m - 1) \geq k$, and the induced law is Geom(1/2) with mean 2, independently of a and r.

Remark 2.5 (What it does *not* give)**.** Corollary 2.3 concerns *uniformly distributed* residues. A single deterministic orbit is not a random sample: its valuations are fixed, not drawn. The corollary is therefore the exact form of the folklore heuristic "$\mathbb{E}[\mathrm{v}_2] = 2$, hence expected factor $3/4 < 1$ per step", but it implies nothing about any particular orbit. That a negative average drift need not decide a deterministic orbit is shown concretely by the map $n \mapsto \lfloor \sqrt{2}\, n + 4 \rfloor$ followed by halving, which Althöfer and Zipproth [2] prove diverges from every odd start.

2.1 The full statistics of the step valuations

The equidistribution law fixes the first moment. The same elementary machinery pins down the whole joint law of the two valuations $v_+ = \mathrm{v}_2(3n + 1)$ and $v_- = \mathrm{v}_2(3n - 1)$ produced by one accelerated step, over n uniform on the odd residues modulo 2^K. Every statement below is *exact* (a rational, verified as such) and is machine-checked in the sense of Section 6; none uses floating point.

Proposition 2.6 (Rotation law)**.** *For every odd a and all $k \geq 1$,*

$$\mathrm{v}_2(an+1) \geq k \iff n \equiv -a^{-1} \pmod{2^k},$$

a single residue class. Hence $\Pr[\mathrm{v}_2(an+1) = k] = 2^{-k}$*, independently of a: the base does not change the law, it merely* rotates *the class by a^{-1}.*

Proof. $\mathrm{v}_2(an+1) \geq k$ means $an \equiv -1 \pmod{2^k}$, i.e. $n \equiv -a^{-1}$, which exists and is unique because a is a 2-adic unit. One class out of 2^{k-1} odd residues gives the geometric weight; subtracting consecutive levels gives 2^{-k}. □

Proposition 2.7 (Second moment)**.** $\mathbb{E}[v_+^2] = 6 - (4K+6)2^{-K}$*, so* $\mathbb{E}[v_+^2] \to 6$ *and* $\mathrm{Var}(v_+) \to 6 - 2^2 = 2$ *— exactly the mean and variance of* $\mathrm{Geom}(1/2)$*.*

Proposition 2.8 (The two valuations are anti-correlated)**.** $\mathrm{v}_2(3n+1)$ *and* $\mathrm{v}_2(3n-1)$ *satisfy* $\min(v_+, v_-) = 1$ *identically: $3n+1$ and $3n-1$ are consecutive even numbers, so exactly one is* $\equiv 2 \pmod 4$*. Consequently $v_+ v_- = \max(v_+, v_-)$, whence* $\mathbb{E}[v_+ v_-] \to 3$ *and*

$$\mathrm{Cov}(v_+, v_-) = \mathbb{E}[v_+ v_-] - \mathbb{E}[v_+]\,\mathbb{E}[v_-] = 3 - 2 \cdot 2 = -1.$$

Proof. If v_+, v_- were independent the product mean would be 4. Because one factor is always 1, the product equals the other factor, i.e. $\max(v_+, v_-)$; and $\mathbb{E}[\max] = \mathbb{E}[v_+ + v_-] - \mathbb{E}[\min] = 4 - 1 = 3$. Subtracting the product of the means gives -1. □

The anti-correlation is the quantitative face of the up/down dichotomy: whenever one child of the two-sided step $n \mapsto \mathrm{odd_part}(3n \pm 1)$ rises sharply, the other is *forced* to valuation 1. The dichotomy is usually stated qualitatively; here it has a number, Cov $= -1$.

3 Which bases admit closed-form orbits

Fix an odd $a \geq 3$ and consider $\mathrm{step}_a(x) = \mathrm{odd_part}(ax+1)$ on odd integers. Every odd x has a unique *block* form

$$x = 2^{r+1}m - 1, \qquad m \text{ odd},\ r \geq 0, \tag{2}$$

in which $r = \mathrm{v}_2(x+1) - 1$.

Theorem 3.1 (One-step law and the characterisation)**.** *Let $a \geq 3$ be odd and write $a - 1 = 2^e q$ with q odd. For every odd m and every $r \geq e$, put $x = 2^{r+1}m - 1$. Then*

$$\mathrm{v}_2(ax+1) = e \qquad \textit{and} \qquad \mathrm{step}_a(x) = a\,2^{r+1-e}m - q. \tag{3}$$

Consequently the closed-form law $\mathrm{step}_a(x) = a\,2^{r+1-e}m - 1$ *holds* if and only if $q = 1$*, i.e. iff $a - 1$ is a power of 2, i.e. iff $a = 2^e + 1$. In that case iteration gives*

$$x_i = a^i\,2^{r+1-ie}m - 1 \qquad \textit{for every } i \textit{ with } ie \leq r. \tag{4}$$

Proof. Compute directly:

$$ax + 1 = a(2^{r+1}m - 1) + 1 = a\,2^{r+1}m - (a-1) = 2^e(a\,2^{r+1-e}m - q).$$

Since $r \geq e$ the bracket is (even)−(odd), hence odd; so $\mathrm{v}_2(ax+1) = e$ *exactly* and $\mathrm{step}_a(x)$ equals the bracket, which is (3). The image equals $a2^{r+1-e}m - 1$ precisely when $q = 1$, which is both directions of the characterisation.

For (4), let $q = 1$. The image $a2^{r+1-e}m - 1$ again has the shape (2), with odd parameter $m' = am$ and exponent $r' + 1 = r + 1 - e$; so the family is preserved and induction on i gives (4) as long as $ie \leq r$. □

Note that (3) holds for *every* odd a — it is the general one-step law — and the characterisation is the observation that its right-hand side lands back in the family exactly when the odd part q of $a-1$ is trivial.

Remark 3.2 (Terminology)**.** The bases $a = 2^e + 1$ are $3, 5, 9, 17, 33, 65, 129, 257, 513, \ldots$ We call them *Fermat-type* because the Fermat numbers $2^{2^k} + 1$ are the sub-family $e = 2^k$; the two sets are *not* equal — 9, 33, 65, 129 and 513 are of the form $2^e + 1$ without being Fermat numbers, and only $3, 5, 17, 257, 65537$ are Fermat primes.

a	$a-1 = 2^e q$	q	closed form?
3	$2^1 \cdot 1$	1	yes (Collatz)
5	$2^2 \cdot 1$	1	yes
7	$2^1 \cdot 3$	3	no
9	$2^3 \cdot 1$	1	yes
11	$2^1 \cdot 5$	5	no
13	$2^2 \cdot 3$	3	no
15	$2^1 \cdot 7$	7	no
17	$2^4 \cdot 1$	1	yes
33	$2^5 \cdot 1$	1	yes

Table 1: The closed form holds exactly when the odd part q of $a-1$ is 1 (Theorem 3.1). The one-step law (3) and the closed form were checked independently of each other — so the "no" column is a genuine test, not a restatement of the hypothesis — for all odd $a \le 999$, $m \le 200$ odd and 8 consecutive $r \ge e$. The bases with $q = 1$ up to 999 are $3, 5, 9, 17, 33, 65, 129, 257, 513$.

Corollary 3.3. *Collatz ($a = 3$) is the smallest Fermat-type base; the next are $5, 9, 17, 33, \ldots$ (all such bases are odd, $2^e + 1$ being odd for $e \ge 1$). The heuristic drift of* step_a *is* $\lambda_a = \log a - \mathbb{E}[\mathrm{v}_2]\log 2 = \log a - 2\log 2$ *(by Corollary 2.3), negative iff $a < 4$. Hence $a = 3$ is the unique odd base on the contracting side, and every odd $a \ge 5$ is super-critical.*

In view of Remark 2.5, the corollary is a statement about averages and decides no individual orbit; in particular it says nothing about whether $9x + 1$ diverges from $x = 1$.

4 A transcendental density constant

Counting the odd integers whose first block does not immediately shrink leads to

$$C_{\mathrm{Collatz}} = \frac{1}{2} - \sum_{n \ge 2} 2^{-\lfloor n \log_2 3 \rfloor}. \tag{5}$$

Summing (5) to $n = 300$ in 80-digit arithmetic (the tail is below 2^{-475}):

$$C_{\mathrm{Collatz}} = 0.28627450232410874196\ldots$$

Lemma 4.1 (Beatty bookkeeping)**.** *For every $n \in \mathbb{N}$, $n + \lfloor n \log_2(3/2) \rfloor = \lfloor n \log_2 3 \rfloor$.*

Proof. $\log_2(3/2) = \log_2 3 - 1$ and $\lfloor y - k \rfloor = \lfloor y \rfloor - k$ for $k \in \mathbb{Z}$. □

The identity is elementary; we record it only because it identifies the two Beatty sequences that occur, and because the equivalent closed form $1 - \sum_{n \ge 1} 2^{-(n + \lfloor n \log_2(3/2) \rfloor)} = C_{\mathrm{Collatz}}$ is the shape that was formalised.

Set $\beta = \log_2 3 \in (1, 2)$, irrational, with Beatty sequence $B = (\lfloor n\beta \rfloor)_{n \ge 1}$, and gap word $w = (w_n)_{n \ge 1}$, $w_n = \lfloor n\beta \rfloor - \lfloor (n-1)\beta \rfloor$.

Lemma 4.2. *$w_n \in \{1, 2\}$ for all n.*

Proof. $\lfloor y + \beta \rfloor - \lfloor y \rfloor \in \{\lfloor \beta \rfloor, \lfloor \beta \rfloor + 1\} = \{1, 2\}$ since $1 < \beta < 2$. □

Lemma 4.3 (Sturmian)**.** *w is Sturmian: its factor complexity is $p(k) = k + 1$ for every $k \geq 1$.*

Proof. By Lemma 4.2, w is the difference word of a Beatty sequence of irrational slope β; such words are precisely the mechanical words of slope $\beta - \lfloor \beta \rfloor$, which are Sturmian exactly because the slope is irrational (Morse–Hedlund [8]; see Lothaire [6], Ch. 2). □

Computing the factors of w directly for $n \leq 4000$ returns alphabet $\{1, 2\}$ and $p(k) = k + 1$ for all $k \leq 8$, as predicted.

Since the exponents in (5) increase by $w_n \in \{1, 2\}$, the binary digits of the series are supported exactly on the Beatty set $\{\lfloor n\beta \rfloor\}$; the digit word of C_{Collatz} is the characteristic word of that set, Sturmian by Lemma 4.3.

Theorem 4.4. *C_{Collatz} is transcendental.*

Proof. The binary expansion of C_{Collatz} is Sturmian, hence aperiodic (so C_{Collatz} is irrational) with factor complexity $p(k) = k + 1$. A real number whose base-b expansion is Sturmian is transcendental by Ferenczi–Mauduit [3]; alternatively, the complexity criterion of Adamczewski–Bugeaud [1] shows that an algebraic irrational cannot have an expansion of complexity $p(k) = k + 1$. □

Corollary 4.5. *For every real $a > 1$ with $\log_2 a \notin \mathbb{Q}$, the constant $C_a = 1 - \sum_{n \geq 1} 2^{-\lfloor n \log_2 a \rfloor}$ is transcendental; in particular for every integer a that is not a power of 2.*

Proof. Identical, with $\beta = \log_2 a$: only irrationality of β is used, the gap alphabet being $\{\lfloor \beta \rfloor, \lfloor \beta \rfloor + 1\}$. □

Remark 4.6. The continued fraction of C_{Collatz} begins

$$[0;\, 3,\, 2,\, 36,\, 2056,\, 70368878395648,\, \ldots],$$

the large early partial quotients reflecting exceptionally good dyadic-rational approximations; only terms stable across two independent precisions are reported. We checked C_{Collatz} against the CODATA physical constants and the standard mathematical constants and found no match within 10^{-3}; we know of no closed form for it, and it is transcendental by Theorem 4.4. It is not, however, a constant of a new kind: it belongs to the classical family $\sum_{n \geq 1} b^{-\lfloor n\alpha \rfloor}$, studied since Böhmer [12], who proved transcendence when α has unbounded partial quotients, and later by Danilov [13], Davison [14] and Adams–Davison [15], the last of whom give the continued fraction of $(b-1)\sum_{n \geq 1} b^{-\lfloor n\alpha \rfloor}$ explicitly for every irrational α. What is specific here is only that this member of the family, with $b = 2$ and $\alpha = \log_2 3$, is the one the Collatz observable produces.

5 No finite-residue reduction

Experimentation with (1) makes it tempting to conclude that the step is determined by a residue of x — for instance modulo 24, where the residue *class* does evolve deterministically. It is not, and we state this explicitly because the error recurs.

Proposition 5.1. *For every modulus M, the valuation $\nu_2(3x+1)$ is* not *a function of $x \bmod M$.*

Proof. Write $M = 2^s M'$ with M' odd, and let c be the unique solution of $3c \equiv -1 \pmod{2^{s+1}}$; it is odd, since $c = -3^{-1}$ and 3^{-1} is a unit. Every $x \equiv c \pmod{2^s}$ has $\nu_2(3x+1) \geq s$, but the exact value is decided by the binary digits of x above position s, which the residue $x \bmod M$ does not see. Concretely, by the Chinese Remainder Theorem choose odd x, y with $x \equiv y \equiv c \pmod{2^s}$

and $x \equiv y \pmod{M'}$ — so $x \equiv y \pmod{M}$ — while additionally $3x+1 \equiv 2^s \pmod{2^{s+1}}$ and $3y+1 \equiv 0 \pmod{2^{s+1}}$; the two congruence systems are compatible because they constrain x and y modulo 2^{s+1} and M' at coprime places. Then $\mathrm{v}_2(3x+1) = s$ while $\mathrm{v}_2(3y+1) \geq s+1$. □

The point is that $\mathrm{v}_2(3x+1) \geq s$ is decided by $x \bmod 2^s$, but $\mathrm{v}_2(3x+1) = s$ exactly is not: the valuation sequence needs unbounded 2-adic precision, so the dynamics is *not* a finite automaton on residues.

The case $M = 24$, explicitly. Among the 2000 odd $x < 4000$:

- 500 (25%) have $\mathrm{v}_2(3x+1) \neq \mathrm{v}_2(3(x+24)+1)$; the smallest is $x = 5$, $y = 29$, with $\mathrm{v}_2(16) = 4 \neq 3 = \mathrm{v}_2(88)$.
- 1959 (98%) have $\mathrm{step}(x) \not\equiv \mathrm{step}(x+24) \pmod{24}$ — so even the residue class of the image is not determined. The smallest witness is $x = 1$, $y = 25$: $\mathrm{step}(1) = 1$ and $\mathrm{step}(25) = 19$, and $1 \not\equiv 19 \pmod{24}$.

An argument reducing the dynamics to finitely many residue classes therefore cannot be correct. We stress a related trap: verifying a claimed identity on the M residue *representatives* $x = 1, 3, \ldots, M-1$ establishes nothing about the other members of each class — $x = 1$ passes while $x = 25$, in the same class, fails.

6 Verification and formalisation

Result	Status	Evidence
Thm 2.2	proved; machine-checked	Lean 4: `geometric_distribution_informal`
Cor. 2.3	proved (counting)	from Thm 2.2; recomputed exactly
Prop. 2.6	proved; machine-checked	Lean 4: rotation law, 6 bases certified
Prop. 2.7	proved	$\mathbb{E}[v_+^2] = 6 - (4K+6)2^{-K}$ exact, $K \leq 18$
Prop. 2.8	proved	$\mathrm{Cov} = -1$ exact; $v_+ v_- = \max$ (0 exc.)
Thm 3.1	proved; machine-checked (both ways)	Lean 4: `closed_form_iff_fermat`
Lem. 4.1	proved (elementary)	one line; also in Lean
Lem. 4.3	proved (Morse–Hedlund)	$p(k) = k+1$ recomputed for $k \leq 8$
Thm 4.4	proved via [3, 1]	C_{Collatz} to 80 digits; expansion Sturmian
Prop. 5.1	proved	1959/2000 witnesses at $M = 24$, $x < 4000$

Table 2: Status of each result. "Machine-checked" means Lean's kernel accepts the proof with no `sorry` and only the ambient axioms `propext`, `Classical.choice`, `Quot.sound`.

The Lean development builds with `lake build CollatzProof.KodamaCollatz CollatzProof.KodamaFermat`, and `#print axioms` is inspected for every theorem cited above; all report exactly `[propext, Classical.choice, Quot.sound]`. The audit is kept as `CollatzProof/AxiomAudit.lean` so that it re-runs on every build. Both halves of Theorem 3.1 are in the kernel: `step_a_general` is the one-step law (3) for every odd base, and `closed_form_iff_fermat` is the biconditional itself. The numerical statements (the value of C_{Collatz}, the gap alphabet, the factor complexity, Table 1, the witness counts) are regenerated by a single script using exact integer and high-precision decimal arithmetic — no floating point is used where exactness is claimed.

7 What is not claimed

- No progress on the Collatz conjecture. Sections 2–4 are independent of it, and Section 5 is limitative.
- No claim of novelty for Lemma 4.1, which is one line.

- Corollary 2.3 is about uniform residues, not about individual orbits (Remark 2.5).

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